

Rigorous Formulation and Modular Reduction of the Erdős-Straus Conjecture

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Abstract

This document presents a formal axiomatic definition, contextual literature review, rigorous modular decompositions, and a structured autoformalization architecture in Lean 4 for the Erdős-Straus Conjecture. It explicitly proves polynomial parameterizations for specific congruence classes to satisfy zero-ellipse constraints.

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1 Axiomatic Definitions

Definition 1.1. Let $\mathbb{N}$ denote the set of non-negative integers, and $\mathbb{N}_{>0}$ the set of strictly positive integers. A rational number $q \in \mathbb{Q}$ is said to have a 3-unit fraction decomposition if there exist $x, y, z \in \mathbb{N}_{>0}$ such that:

$$q = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

Definition 1.2. An integer $n \in \mathbb{N}$ with $n \geq 2$ satisfies the Erdős-Straus property if the rational number $\frac{4}{n}$ admits a 3-unit fraction decomposition. Formally:

$$\text{SatisfiesErdosStraus}(n) \iff \exists x, y, z \in \mathbb{N}_{>0}, \frac{4}{n} = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

2 Contextual Literature

The Erdős-Straus conjecture postulates that for all $n \geq 2$, $\text{SatisfiesErdosStraus}(n)$ holds true. While unproven for all n , the problem is deeply connected to Diophantine equations and the Egyptian fraction representation of rational numbers.

Recent literature, such as “A Complete Congruence System for the Erdos-Straus Conjecture” and “Structure and form of the solutions of the Erdos-Straus conjecture”, attacks the problem by analyzing the structure of its solutions modulo specific primes. A common technique involves establishing polynomial identities that provide explicit values for x, y, z in terms of n when n belongs to certain congruence classes modulo k , where k is a carefully chosen integer.

This approach is analogous to the use of covering systems of congruences, pioneered by Erdős himself to construct odd integers that are not of the form $p + 2^k$.

3 Lemma Isolation and Strategy

The global conjecture can be reduced to proving the property for prime numbers. If $n = p \cdot q$, and $4/p = 1/x + 1/y + 1/z$, then $4/n = 1/(qx) + 1/(qy) + 1/(qz)$. We structure our strategy by covering the integers with congruence classes modulo 24. We isolate a specific lemma to explicitly demonstrate the zero-ellipse proof methodology for a non-trivial congruence class.

Lemma 3.1. Let $n \in \mathbb{N}_{>0}$. If $n \equiv 2 \pmod{3}$, then $\text{SatisfiesErdosStraus}(n)$ is true.

4 Rigorous Proof (Zero-Ellipse)

Proof. Let n be an integer strictly greater than 1. We assume that $n \equiv 2 \pmod{3}$. By the definition of modular congruence, there exists a non-negative integer $k \in \mathbb{N}$ such that:

$$n = 3k + 2$$

We propose the following explicit parameterization for the variables x, y , and z :

$$x = n = 3k + 2$$

$$y = k + 1$$

$$z = n(k + 1) = (3k + 2)(k + 1)$$

First, we verify that x, y, z are strictly positive integers. Since $k \geq 0$, we have $x = 3k + 2 \geq 2 > 0$. Thus $x \in \mathbb{N}_{>0}$. Similarly, $y = k + 1 \geq 1 > 0$, so $y \in \mathbb{N}_{>0}$. Finally, z is the product of two strictly positive integers, so $z \in \mathbb{N}_{>0}$.

Now, we evaluate the sum of the reciprocal of these variables:

$$S = \frac{1}{x} + \frac{1}{y} + \frac{1}{z}$$

Substituting the proposed values:

$$S = \frac{1}{3k + 2} + \frac{1}{k + 1} + \frac{1}{(3k + 2)(k + 1)}$$

The common denominator for these three fractions is $(3k + 2)(k + 1)$. We convert each fraction to have this common denominator. For the first fraction, we multiply the numerator and denominator by $(k + 1)$:

$$\frac{1}{3k + 2} = \frac{1 \cdot (k + 1)}{(3k + 2)(k + 1)} = \frac{k + 1}{(3k + 2)(k + 1)}$$

For the second fraction, we multiply the numerator and denominator by $(3k + 2)$:

$$\frac{1}{k + 1} = \frac{1 \cdot (3k + 2)}{(k + 1)(3k + 2)} = \frac{3k + 2}{(3k + 2)(k + 1)}$$

The third fraction already has the common denominator:

$$\frac{1}{(3k + 2)(k + 1)}$$

We now sum the three fractions by adding their numerators:

$$S = \frac{(k + 1) + (3k + 2) + 1}{(3k + 2)(k + 1)}$$

We group the k terms and the constant terms in the numerator:

$$S = \frac{(k + 3k) + (1 + 2 + 1)}{(3k + 2)(k + 1)}$$

$$S = \frac{4k + 4}{(3k + 2)(k + 1)}$$

We factor out the constant 4 from the numerator:

$$S = \frac{4(k + 1)}{(3k + 2)(k + 1)}$$

Since $k \geq 0$, we have $k + 1 \neq 0$. We can therefore simplify the fraction by canceling the common factor $(k + 1)$ in the numerator and denominator:

$$S = \frac{4}{3k + 2}$$

Substituting back $n = 3k + 2$, we obtain:

$$S = \frac{4}{n}$$

We have found $x, y, z \in \mathbb{N}_{>0}$ such that $\frac{1}{x} + \frac{1}{y} + \frac{1}{z} = \frac{4}{n}$. This explicitly demonstrates that $\text{SatisfiesErdosStraus}(n)$ is true for all $n \equiv 2 \pmod{3}$. $\square$

5 Architecture for Autoformalization

To structure this problem in Lean 4, we define the required types for unit fractions and the conjecture statement.

```
import Mathlib.Data.Rat.Basic
import Mathlib.Data.Nat.Basic
import Mathlib.Data.Nat.Parity

open Nat

-- Definition of the 3-unit fraction decomposition property
def SatisfiesErdosStraus (n : Nat) : Prop :=
  Exists (fun (x : Nat) =>
    Exists (fun (y : Nat) =>
      Exists (fun (z : Nat) =>
        x > 0 /\ y > 0 /\ z > 0 /\
        (4 : Rat) / (n : Rat) = (1 : Rat) / x + (1 : Rat) / y + (1
          : Rat) / z)))

-- Main Conjecture Statement
theorem erdos_straus_conjecture (n : Nat) (hn : n ≥ 2) :
  SatisfiesErdosStraus n := by
  admit

-- Lemma for n = 2 mod 3
lemma erdos_straus_mod3_2 (n : Nat) (hn : n ≥ 2) (hmod : n % 3 =
  2) :
  SatisfiesErdosStraus n := by
  admit
```